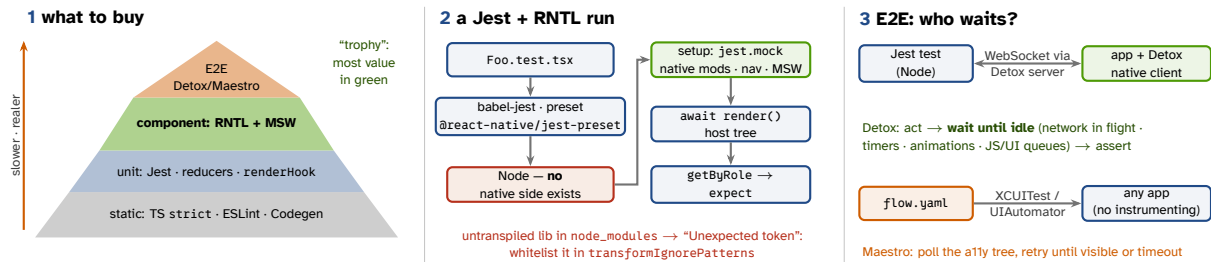


In one line: Buy **confidence** per **second**: TS strict for free, Jest for logic, the **bulk** in **RNTL** component tests that query *like a user* against mocked network, and a **few** E2E journeys — **Detox** (grey-box) or **Maestro** (black-box).



The tools

- Jest** runs in **Node**: the preset (`@react-native/jest-preset` since 0.85) stubs core native bits; every *third-party* native module is `jest.mocked` (many ship a mock, e.g. `react-native-gesture-handler/jestSetup`). Fake timers: `jest.useFakeTimers() + advanceTimersByTime`.
- RNTL** renders the React tree (no pixels, no layout), queries *host* elements: `ByRole('button', {name}) > ByLabelText > ByText > ByTestId` (last resort). Role needs an *accessible* element — tests force real a11y. `getBy` throws · `queryBy` → null (absence) · `findBy` = async wait.
- RNTL 14** (React 19 only): `render`, `renderHook`, `fireEvent`, `act` are **async** — await them (codemod `rntl-v14-async-functions`). `fireEvent.press` just calls the nearest `onPress`; `userEvent` replays `pressIn/Out`, `focus`, per-key `changeText`, `blur`, and respects `disabled / editable={false}`.
- Hooks**: `await renderHook(fn, {wrapper})` → `result.current`; or test *through* the component.
- Network**: `MSW setupServer (msw/node)` — real client + parsing, per-test `server.use()` for errors. Cheaper seam: mock the API module. TanStack Query: **new QueryClient per test**, `retry=false`.
- Navigation**: real `NavigationContainer` + navigator (integration), or mock `useNavigation` and `assertNavigate` (unit). `TurboModule`: mock its `Native<X>.ts` spec file.
- Detox** (Wix): instrumented build, by `.id(testID)`, no sleeps; costs native builds + upgrade breakage; a never-idle app (looping animation, polling) hangs it.
- Maestro**: YAML (`tapOn`, `assertVisible`), no app changes, also native/Flutter; slower, fuzzier waits.

Example — a user-level test

```
it('shows the error, then retries', async () => {
  server.use(http.get('/me', () => HttpResponse.error()));
  const user = userEvent.setup();
  await render(<Profile />, { wrapper: Providers }); // RNTL 14
  expect(await screen.findByText(/could not load/i))
    .toBeOnTheScreen();
  server.resetHandlers(); // next call succeeds
  await user.press(screen.getByRole('button', { name: 'Retry' }));
  expect(await screen.findByText('Ada')).toBeOnTheScreen();
});
```

TypeScript for RN

```
type Load<T> = { status: 'idle' } | { status: 'loading' }
  | { status: 'ok'; data: T } | { status: 'err'; error: Error };
type Props = PressableProps & { title: string;
  style?: StyleProp<ViewStyle>; children?: React.ReactNode };
function List<T>(p: { items: readonly T[];
  key: (t: T) => string; row: (t: T) => React.ReactElement }) {}
type Root = { Home: undefined; Profile: { userId: string } };
type P = NativeStackScreenProps<Root, 'Profile'>;
const gap = { s: 4, m: 8 } satisfies Record<string, number>;
```

```
// NativeCrypto.ts -- Codegen reads this spec
export interface Spec extends TurboModule {
  hash(input: string): Promise<string>; }
export default
  TurboModuleRegistry.getEnforcing<Spec>('NativeCrypto');
```

- Unions**: no loading + error at once; `const x: never = s` in the default = exhaustive.
- Skip `React.FC` (implicit children dropped in React 18 types); extend `host props` via `ComponentProps<typeof X>`.
- `satisfies` (TS 4.9) checks *without widening* — literals and keys stay precise; as just silences the checker.
- `Nav`: global `ReactNavigation.RootParamList` types `useNavigation()`; deep-link params arrive as *strings* — validate.
- `Flags`: `strict`, `noUncheckedIndexedAccess (arr[i]: T | undefined)`, `exactOptionalPropertyTypes`.
- Codegen**: `Native<Name>.ts` + `codegenConfig` (`package.json`) → C++/ObjC++/Java glue: the TS is the ABI, in a subset (number, `Int32`, `string`, `boolean`, `objects`, `arrays`, `Promise`, `callbacks`).

Traps

- ! Types vanish at runtime: `res.json()` as `User` is a lie — validate at the boundary (`zod/valibot`) or type it unknown.
- ! Flakes: un-awaited `findBy/userEvent`; real timers; leaked state (singletons, shared `QueryClient`); live network; E2E permission dialogs.
- ! “not wrapped in `act(...)`” = an update you didn’t wait for: `findBy`, not more `act`.
- ! Whole-screen snapshots = change detectors; asserting state or child internals tests the implementation.
- ! Jest green ≠ works: no native code, layout or threads ran.

Remember

Query like a user · mock at the edge · wait, don’t sleep · type the boundary.